5

Solving Problems By Searching

Chapter Learning Outcomes

CLO

5

- Define a search problem.
- Compute the size of a search tree.
- Compute the branch factor of a search tree.
- Know the branching factors for some games.
- Compute the number of nodes in a search tree.
- Distinguish between maximum depth and goal depth.
- Approximate a time to search for a goal in a search tree.
- Approximate a space when searching for a goal in a search tree.
- Know different search strategies.
- Understand the criteria for evaluating search strategies.
- Understand, apply and evaluate (BFS, DFS, DLS, and IDS) algorithms.

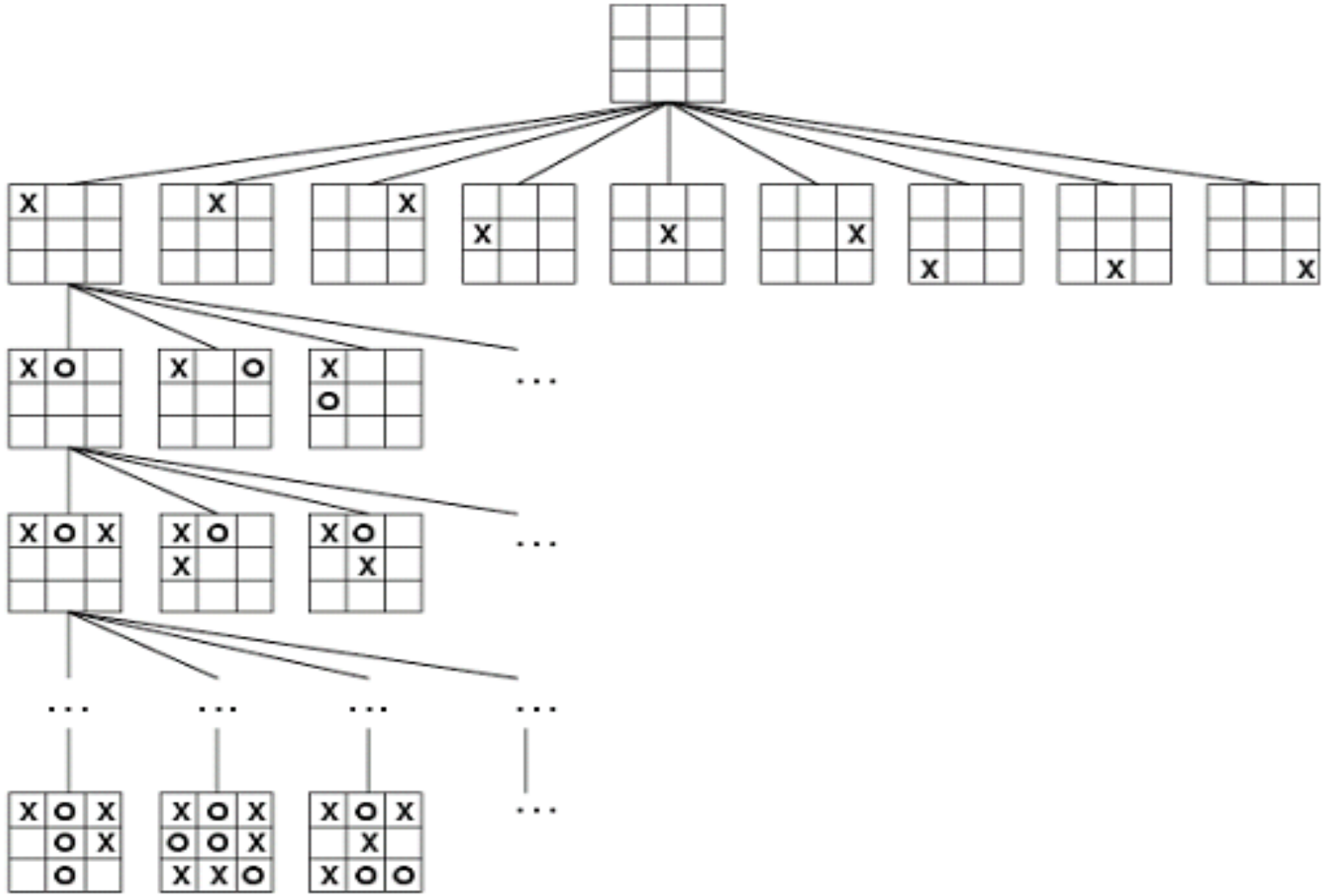
Defining a Search Problem

Search is one of the most powerful approaches to problem solving in AI. A search problem is defined by four items:

- **Initial state**
- **Actions (Operators):** operators are legal actions which the agent can take to move from one state to another.
- **Goal test:** Determine if the current state is the goal state or not.
- **Path cost:**
  - e.g., sum of distances, number of actions executed, etc.

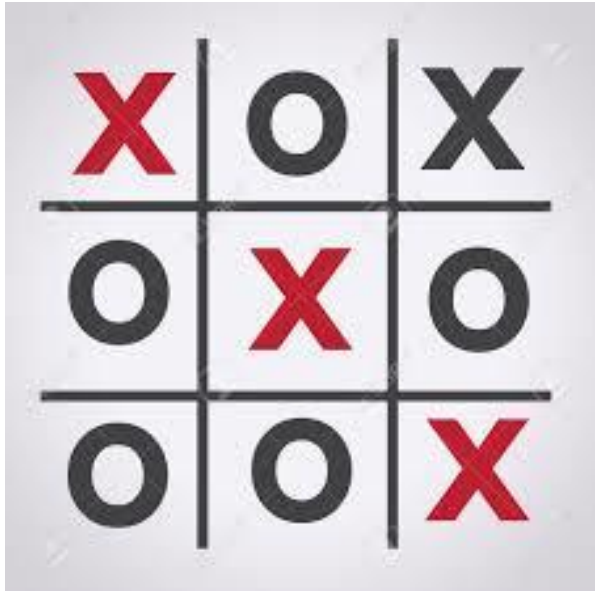
A solution is a sequence of actions leading from the initial state to a goal state

A Search Tree for Tic-Tac-Toe



Quiz

Define the tic-tac-toe game as a search problem in AI.



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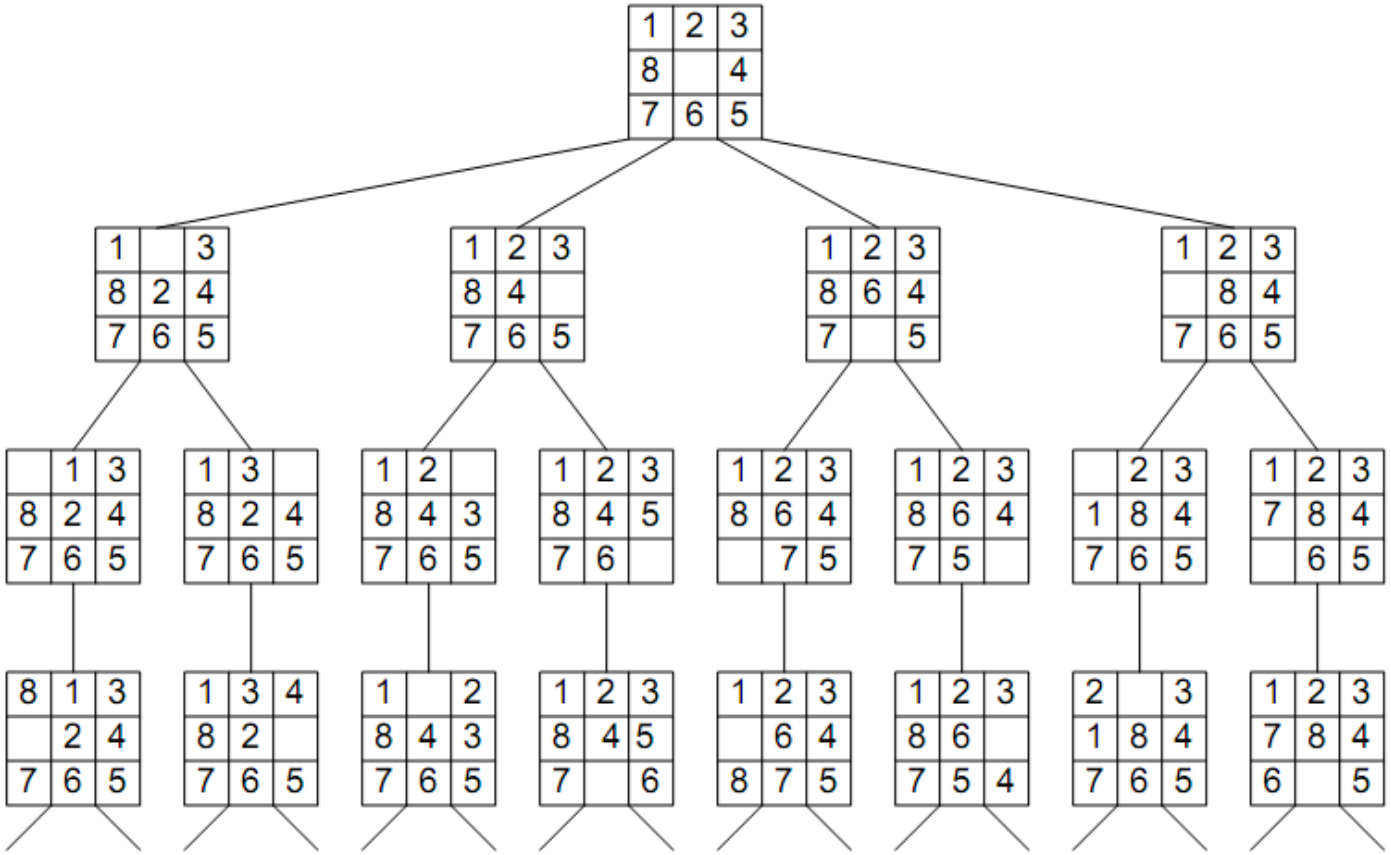
A Search Tree for 8-Puzzle



Typically AI problems will have a very large space.

What is the size of a Search Tree?

8 - Puzzle

2	1	3
4	7	6
5	8	

Number of Nodes = 6,964,551

How did we compute this value?

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What is the size of a Search Tree?

Rubik's Cube

Number of Nodes = 193,484,242,857,296,000,000

How did we compute this value?

What is the size of a Search Tree?

Chess

Number of Nodes =

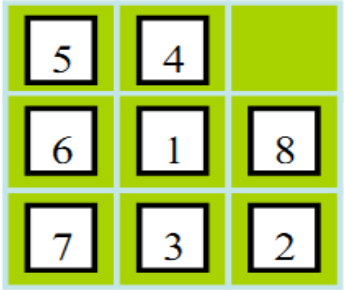
7,833,394,781,046,530,000,000,000,000,000

How did we compute this value?

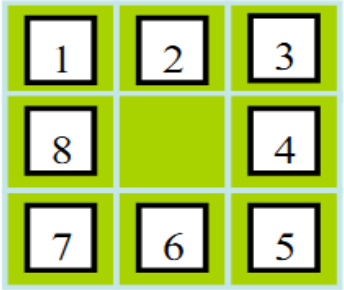
Defining a Search Problem

EXAMPLE

Define the 8-Puzzle game as a search problem in AI.



initial state



goal state

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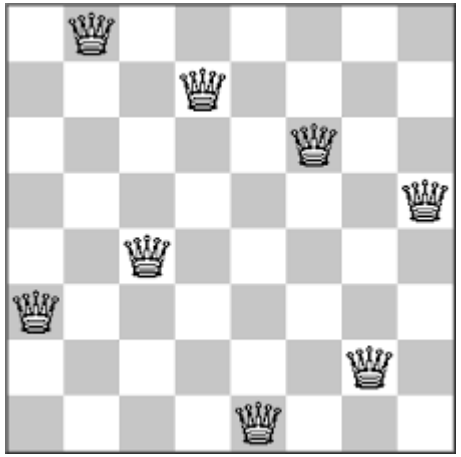
Defining a Search Problem

EXAMPLE

Define the 8-queens game as a search problem in AI.



- **Initial State:** No queens on the board
- **Actions** : add a queen to any square.
- **Goal Test** : 8 queens on board, none attacked.
- **Path cost** : 1 per move

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- Search Tree.
- Branch Factor.
- Number of Sates.

Search Tree

- Each node in a search tree has child nodes that represent each of the states reachable by the parent state.

- Another way of looking at it: child nodes represent all available options once the search procedure reaches the parent node.

- There are a number of strategies for traversing a search tree.

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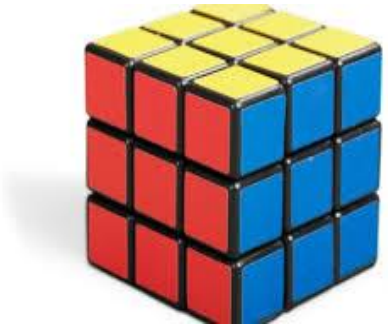
What is the size of a Search Tree?

Number of Nodes = 6,964,551

2	1	3
4	7	6
5	8	

How did we compute these values?

r of Nodes = 193,484,242,857,296,000,000



Number of Nodes =
7,833,394,781,046,530,000,000,000,000,000

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Branch Factor

In computing, tree data structures, and game theory, the **branching factor (BF)** is the number of children at each node. If the branching factor is not uniform, an average branching factor can be calculated.

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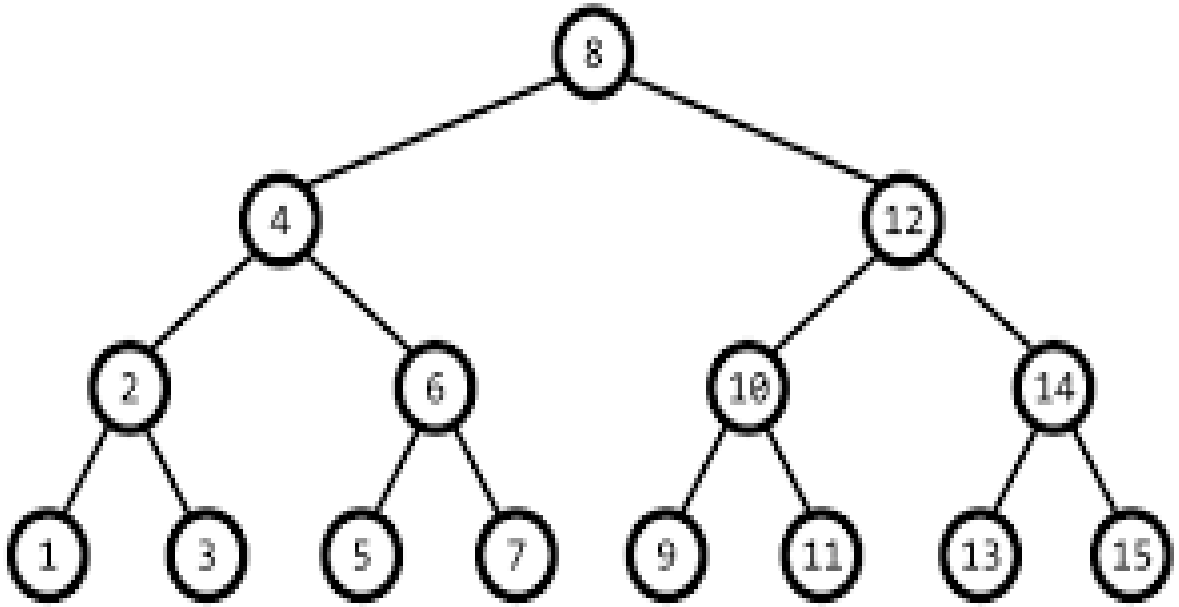
Branch Factor

EXAMPLE

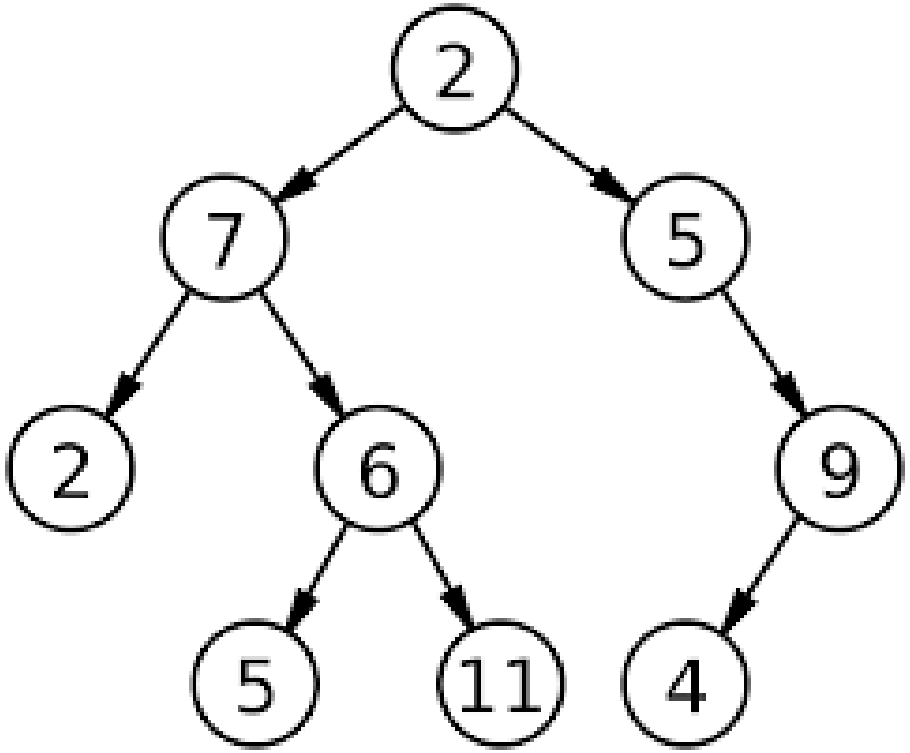
Compute the branch factor (BF) for the following tree:



Branch Factor

EXAMPLE

Compute the branch factor (BF) for the following tree:



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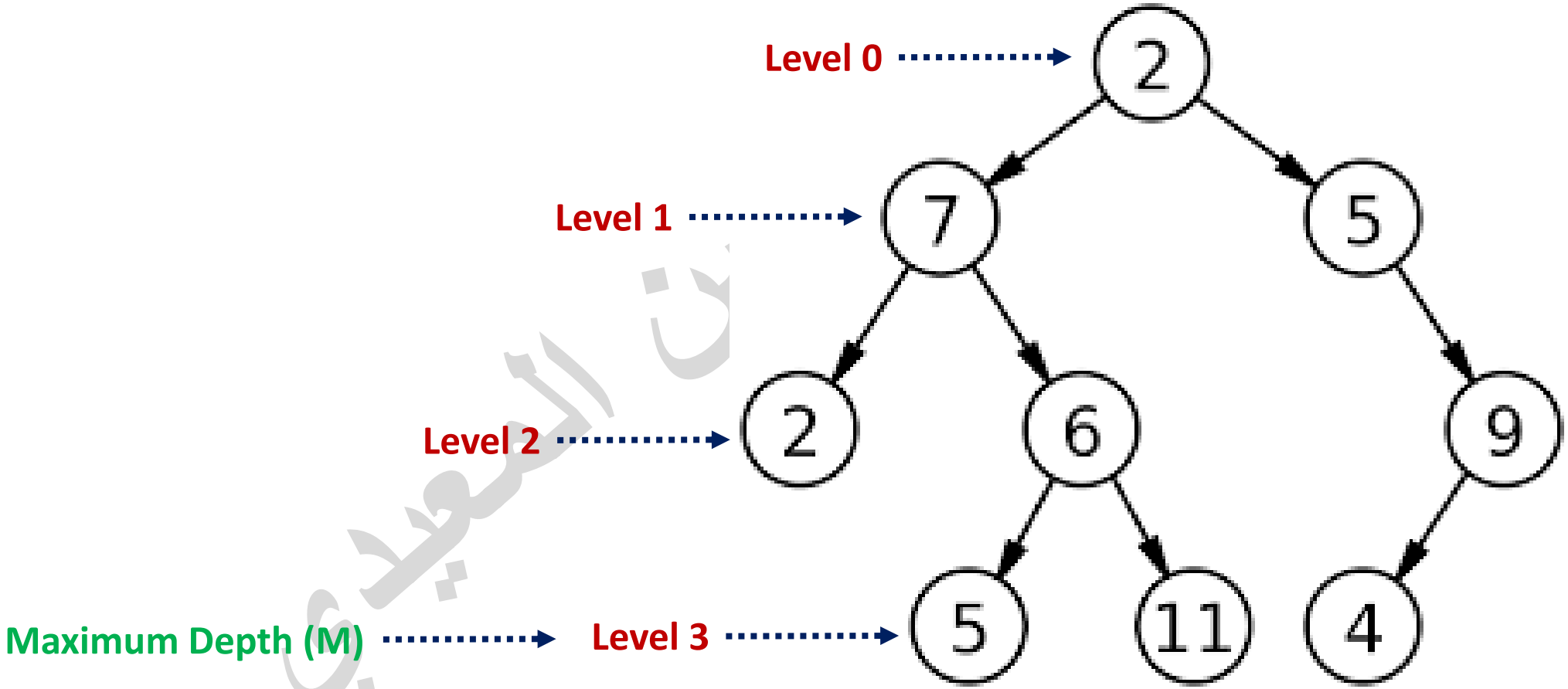
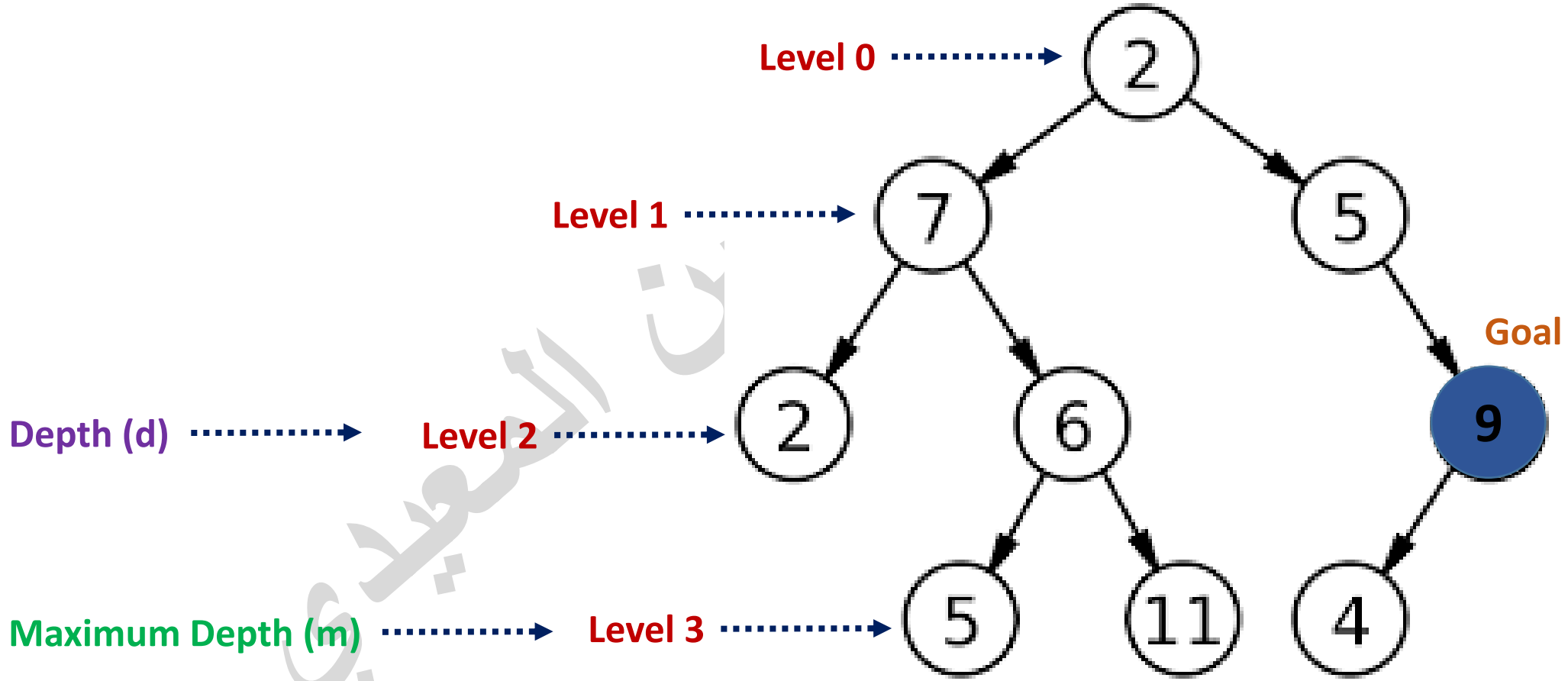
Maximum Depth of a State-Space (Tree)



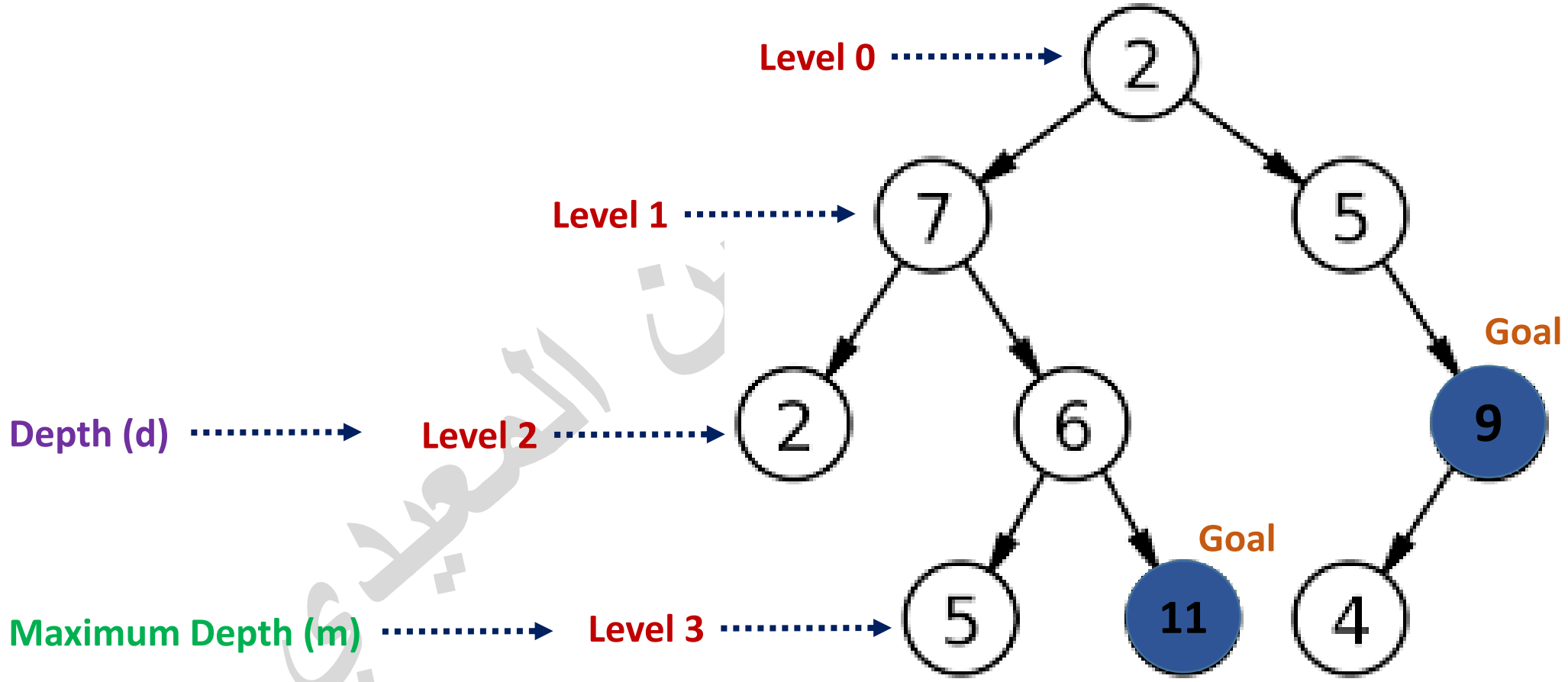
m=3

Depth of the Optimal Solution in a State-Space (Tree)



m=3 d=2

Depth of the Optimal Solution in a State-Space (Tree)



m=3 d=2

How to Find the Number of Nodes in a Search Tree?

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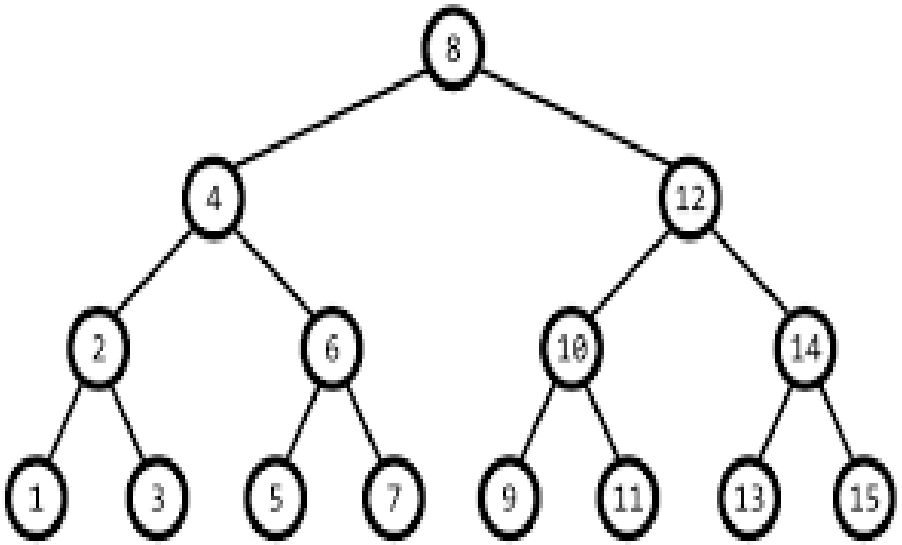
Example

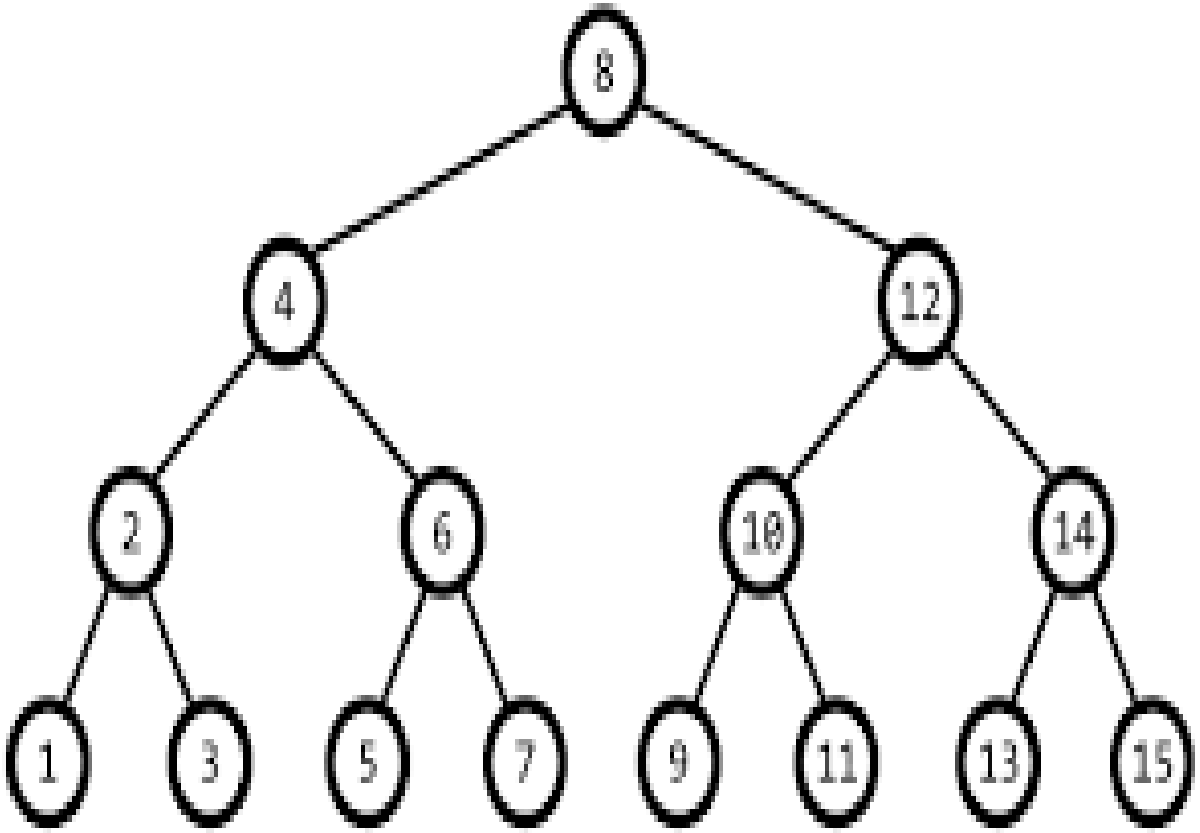
EXAMPLE

Assume the goal state is (15), find:

1. Branch Factor (BF).
2. Maximum Depth (m).
3. Goal Depth (d).
4. Number of Nodes (n).



Example - Solution



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Branching Factors for some Games

8 - Puzzle

2	1	3
4	7	6
5	8	

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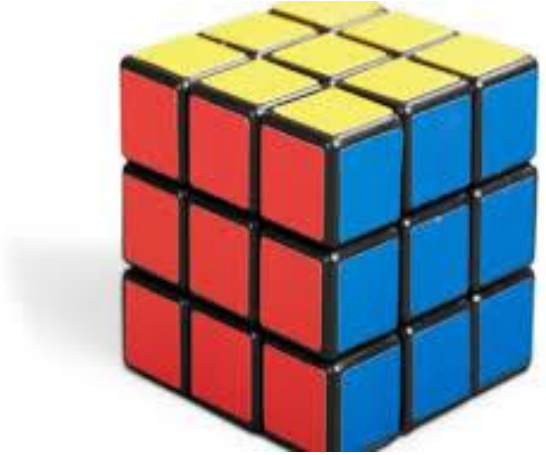
Branching Factors for some Games

Rubik's Cube



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Branching Factors for some Games

Chess

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Example

EXAMPLE

Assume you can process 300 states per second to search a given game tree using a blind search. What is the approximate time to search for a goal, assuming the branching factor (BF)=5 and the goal is in level 7?

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Search Strategies

Un-informed (Blind) Search

- Breadth-First-Search (BFS)
- Depth-First-Search (DFS)
- Depth Limited Search (DLS)
- Iterative Deepening (IDS)
- Bidirectional Search (BS)

Informed (Heuristic) Search

- Greedy Best-First-Search (GBFS)
- A*
- IDA*
- SMA*
- Hill Climbing
- Beam search

Optimization Search

- Hill climbing
- Simulated Annealing
- Genetic Algorithms
- Ant Colony

Game playing

- Minimax Algorithm
- Alpha-Beta Pruning

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Criteria for Evaluating Search Strategies

Completeness	Is the strategy guaranteed to find a solution when there is one?
Optimality	Does the strategy find the highest-quality solution when there are several solutions?
Time Complexity	How long does it take to find a solution?
Space Complexity	How much memory does it need to perform the search?

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Breadth-First Search

(BFS)

Breadth-First Search (BFS)

- BFS belongs to uninformed (blind) strategies (algorithms).

- Uninformed (blind) strategies use only the information available in the problem definition. These strategies order nodes without using any domain specific information.

- The following are examples for uninformed (blind) strategies.
  - **Breadth-First-Search (BFS)**
  - **Depth-First-Search (DFS)**
  - **Depth Limited Search (DLS)**
  - **Iterative Deepening (IDS)**
  - **Bidirectional Search (BS)**

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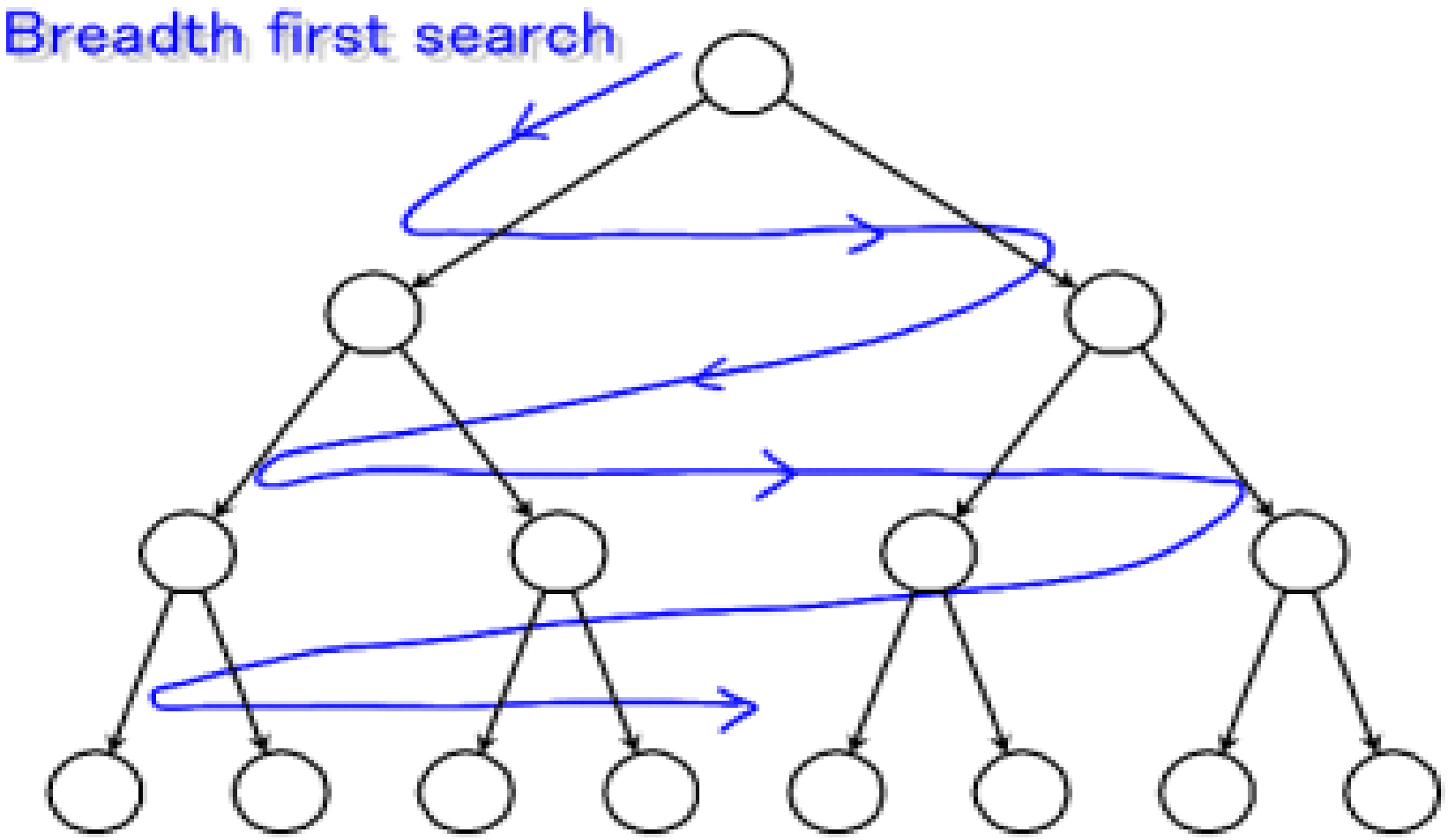
Breadth-First Search (BFS)



Breadth-First Search (BFS)

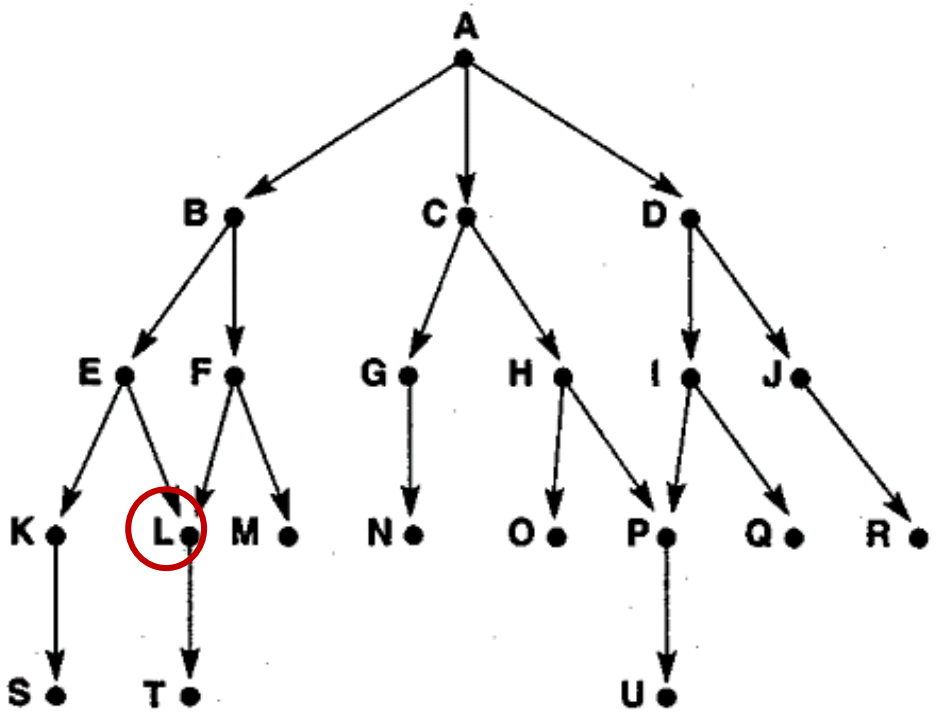
```
function breadth_first_search;

begin
   open := [Start];                                                    % initialize
   closed := [ ];
   while open ≠ [ ] do                                                 % states remain
      begin
         remove leftmost state from open, call it X;
            if X is a goal then return SUCCESS                         % goal found
               else begin
                  generate children of X;
                  put X on closed;
                  discard children of X if already on open or closed;  % loop check
                  put remaining children on right end of open          % queue
               end
      end
   return FAIL                                                         % no states left
end.
```

Breadth-First Search (BFS) - Example

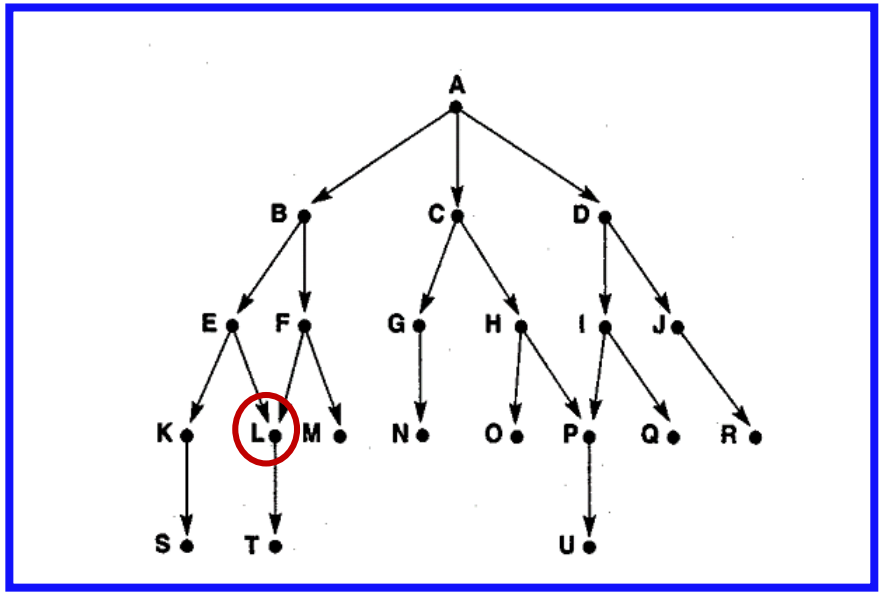
EXAMPLE

Apply the BFS algorithm for the following graph.



Breadth-First Search (BFS)

i	[OPEN]	X
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		



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Breadth-First Search (BFS)

1- What is the content of OPEN array at iteration 3?

2- What is the value of (x) at iteration 3?

3- What is the first item of OPEN array at iteration 3?

4- What is the second item of OPEN array at iteration 3?

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Breadth-First Search (BFS)

5- What is the content of OPEN array at iteration 7?

6- What is the value of (x) at iteration 5?

7- What is the order of the expanded states?

8- What is the solution path?

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Evaluating the Breadth-First Search (BFS)

- One of the simplest search strategy.
- Time and Space are complex.
- Cannot be use to solve any but it can use to solve the smallest problems. See Slide (338) for a simulation.

Completeness — **Is the strategy guaranteed to find a solution when there is one?**

Optimality — **Does the strategy find the highest-quality solution when there are several solutions?**

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Evaluating the Breadth-First Search (BFS)

Time Complexity How long does it take to find a solution?

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Evaluating the Breadth-First Search (BFS)

Space Complexity

How much memory does it need to perform the search?

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Examples of Time and Memory Requirements for (BFS)

Depth of Solution	Nodes Expanded	Time	Memory
0	1	1 millisecond	100 bytes
2	111	0.1 seconds	11 kbytes
4	11,111	11 seconds	1 megabyte
8	10^8	31 hours	11 gigabytes
12	10^{12}	35 years	111 terabytes

Assuming:

- BF=10
- 1000 nodes/sec
- 100 bytes/node

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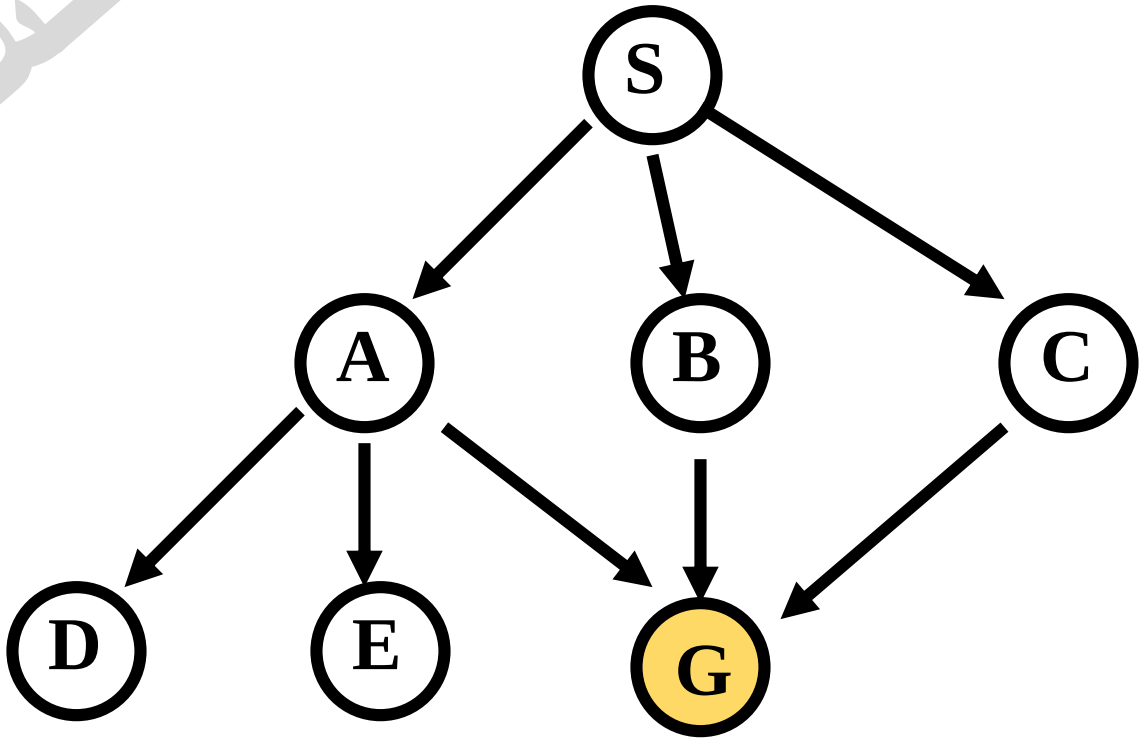
Quiz

Apply the BFS algorithm for the following graph.



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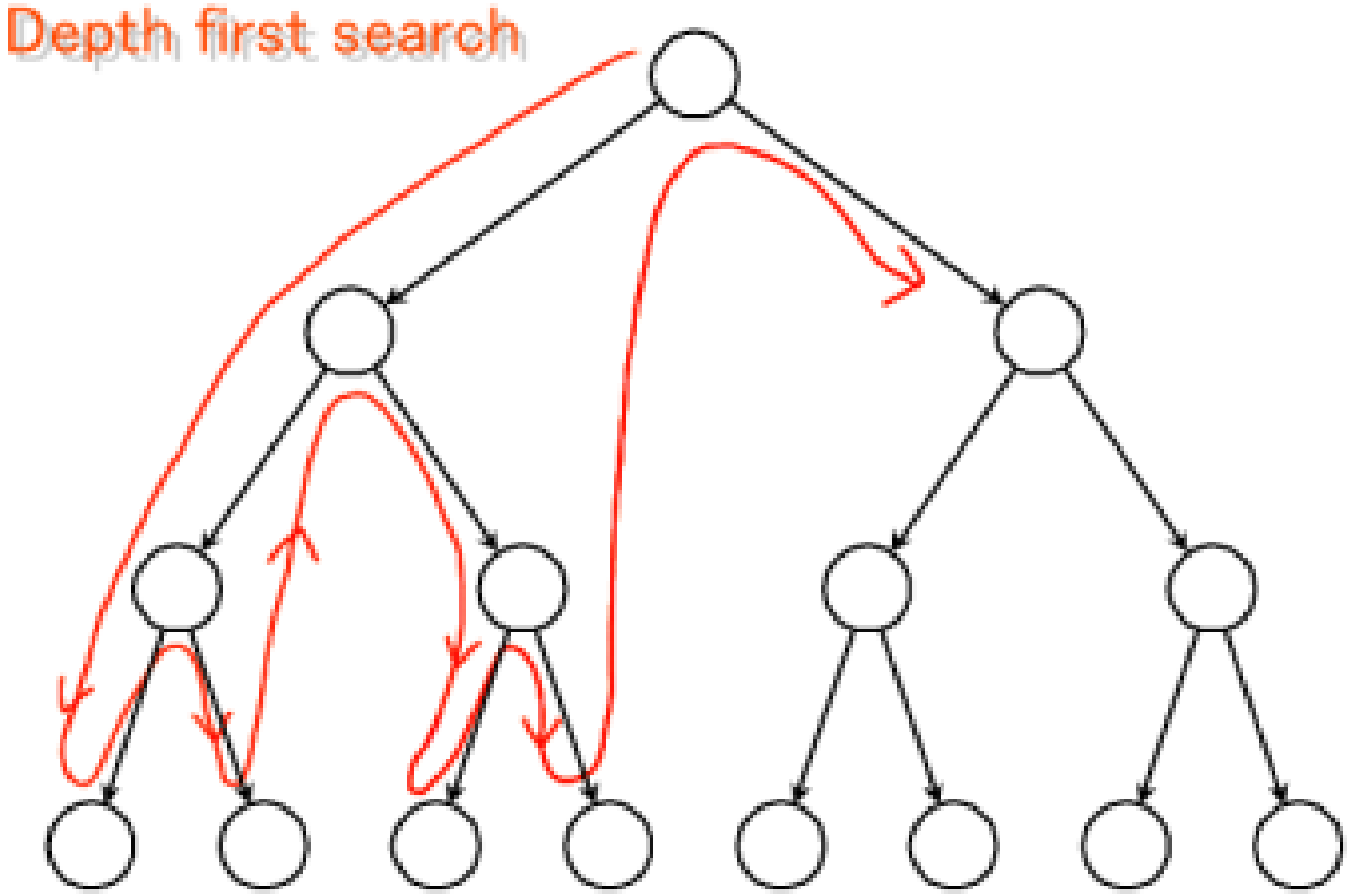
Depth-First Search

(DFS)

Depth-First Search (DFS)



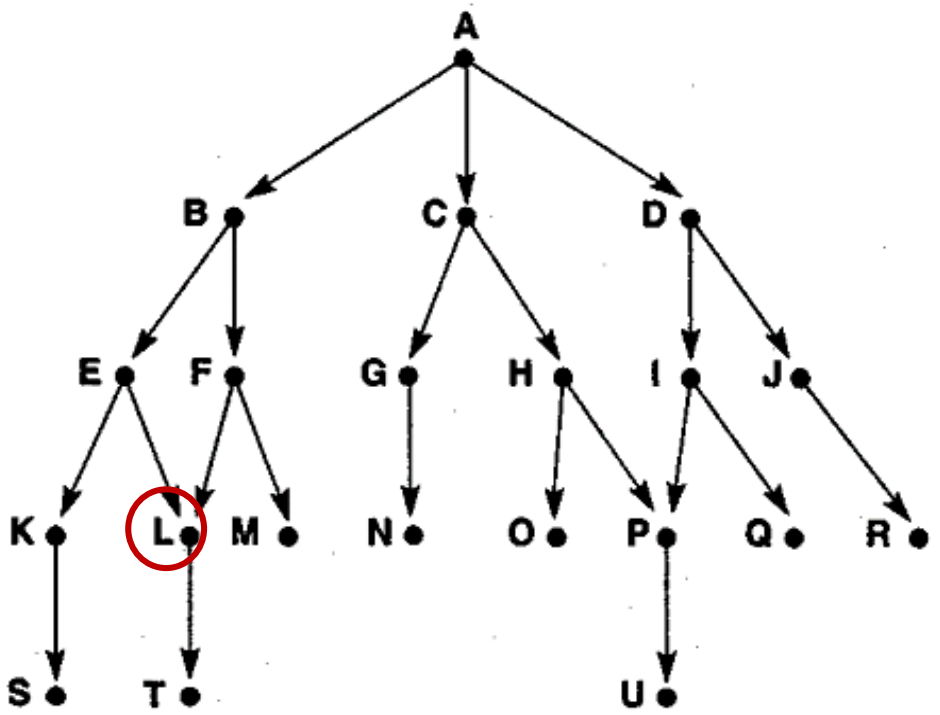
Depth-First Search (DFS)

```
begin
   open := [Start];                                                    % initialize
   closed := [ ];
   while open ≠ [ ] do                                                 % states remain
      begin
         remove leftmost state from open, call it X;
         if X is a goal then return SUCCESS                            % goal found
            else begin
               generate children of X;
               put X on closed;
               discard children of X if already on open or closed;     % loop check
               put remaining children on left end of open              % stack
            end
      end;
   return FAIL                                                         % no states left
end.
```

Depth-First Search (DFS) - Example

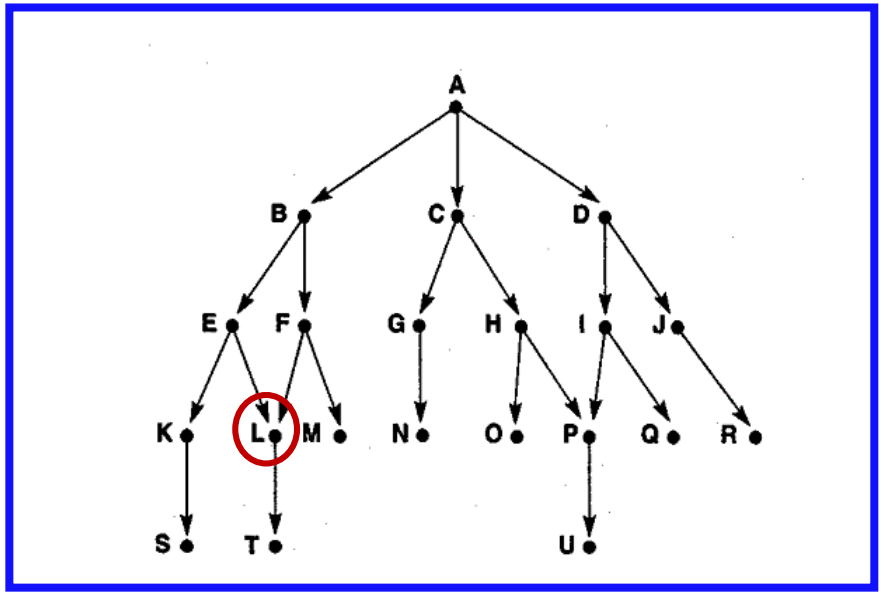
EXAMPLE

Apply the DFS algorithm for the following graph.



Depth-First Search (DFS)

i	[OPEN]	X
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		



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Depth-First Search (DFS)

1- What is the content of OPEN array at iteration 3?

2- What is the value of (x) at iteration 3?

3- What is the first item of OPEN array at iteration 3?

4- What is the second item of OPEN array at iteration 3?

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Depth-First Search (DFS)

5- What is the content of OPEN array at iteration 4?

6- What is the value of (x) at iteration 5?

7- What is the order of the expanded states?

8- What is the solution path?

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Evaluating the Depth-First Search (DFS)

- One problem with DFS is that it can get stuck going down the wrong path.
- Time and Space are complex.
- Many problems have **very deep** or even **infinite** search trees.
- DFS should be **avoided** for search trees with **large or infinite maximum depths**.

Completeness — **Is the strategy guaranteed to find a solution when there is one?**

Optimality — **Does the strategy find the highest-quality solution when there are several solutions?**

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Evaluating the Depth-First Search (DFS)

Time Complexity How long does it take to find a solution?

Space Complexity How much memory does it need to perform the search?

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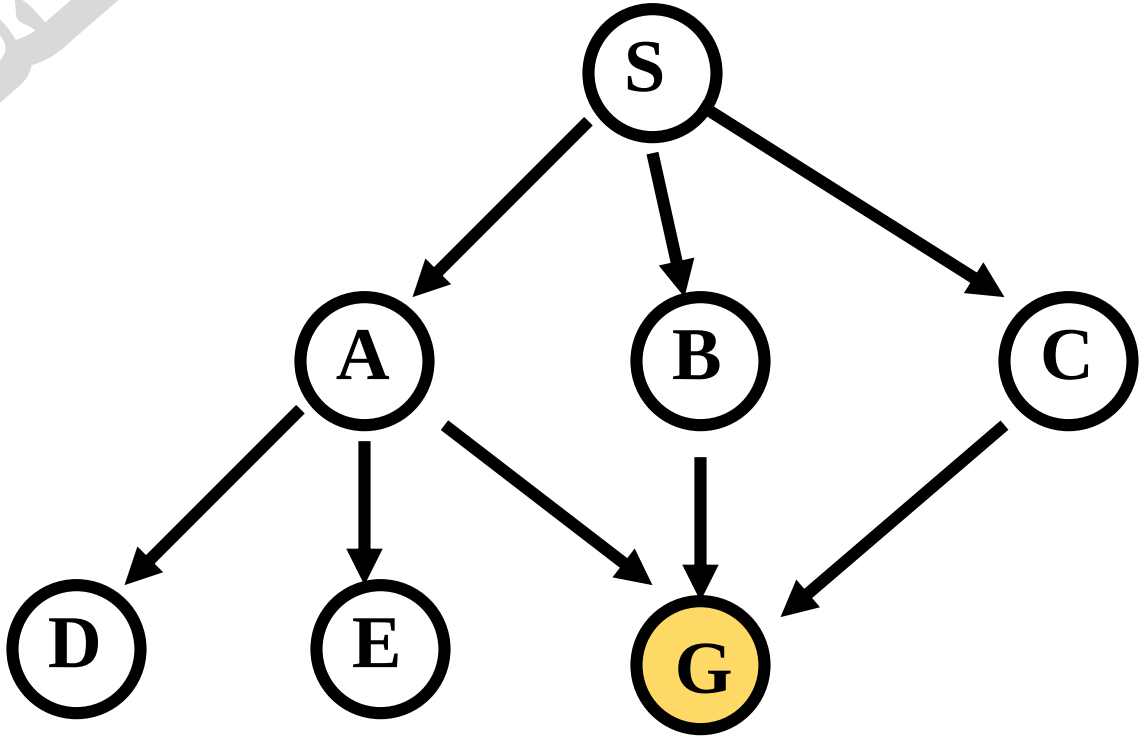
Quiz

Apply the DFS algorithm for the following graph.



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Depth-Limited Search

(DLS)

Depth Limited Search (DLS)

- **Stop DFS at a *fixed* depth *f*.**

- **Goal, may not be found: *Incomplete Algorithm***
  - **If goal depth *d* > *f*.**

- **Why DLS?**
-
- **Time:**

- **Memory:**

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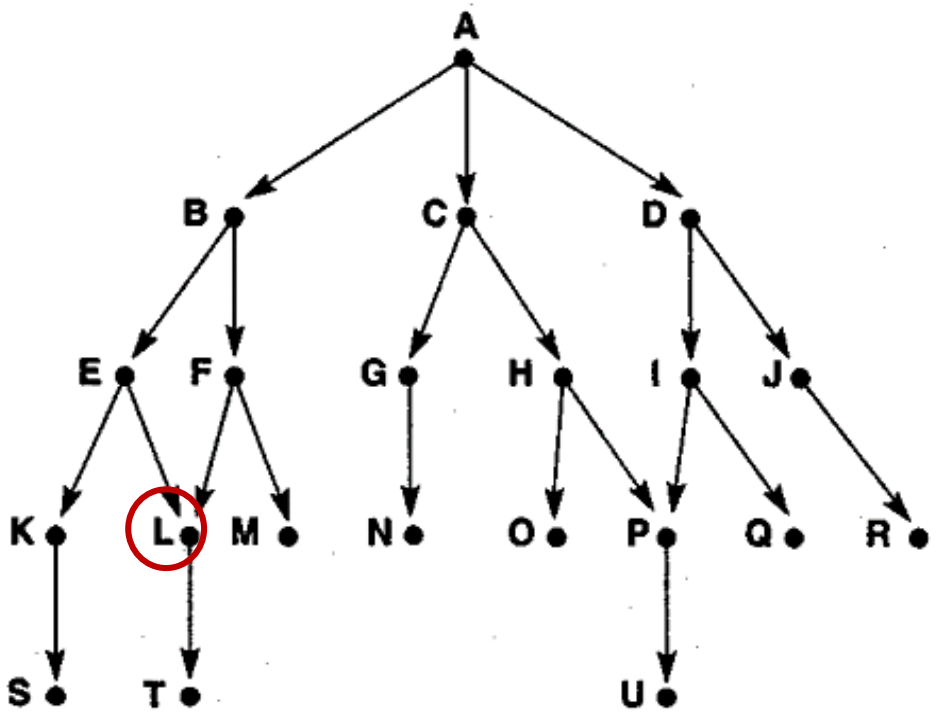
Depth Limited Search (DLS) - Example

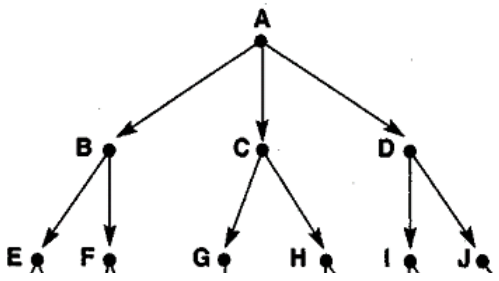
EXAMPLE

Apply the DLS algorithm for the following graph, assume the depth-limit (f) is 2.



Depth Limited Search (DLS) - Example

i	[OPEN]	X
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		



Depth-Limit f = 2

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Iterative Deepening Search (IDS)

Iterative Deepening Search (IDS)

- **Repeatedly stop DFS at a fixed depth *f* (i.e., DLS), AND**
  - **If goal is not found,**
  - **then RESTART from start node again, with f = f+1**

- **Complete Algorithm**
  - **Goal will be found when f = d, goal depth**

- **Time:**

- **Memory:**

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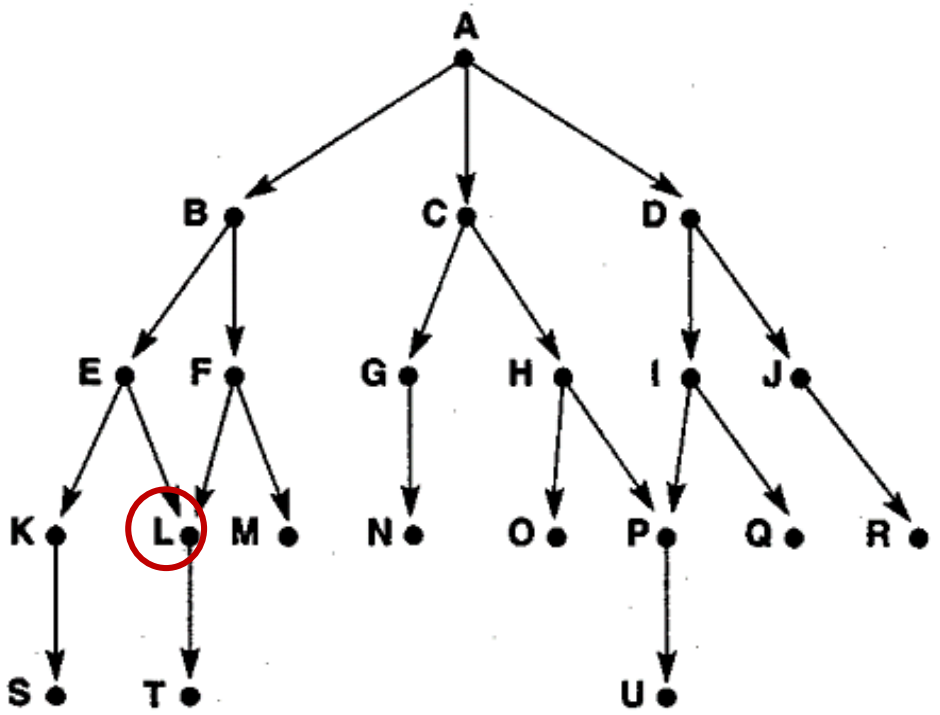
Iterative Deepening Search (IDS) - Example

EXAMPLE

Apply the IDS algorithm for the following graph, start from f =0.



Iterative Deepening Search (IDS) - Example

f = 0

i	[OPEN]	X
1		

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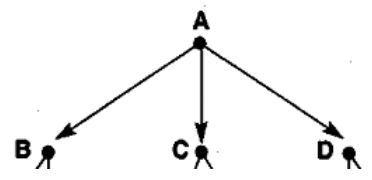
Iterative Deepening Search (IDS) - Example

f = 1

i	[OPEN]	X
1		
2		
3		
4		



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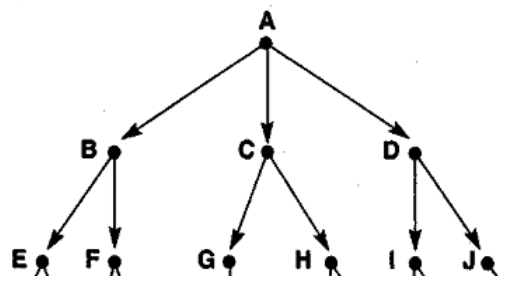
Iterative Deepening Search (IDS) - Example

f = 2

i	[OPEN]	X
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		



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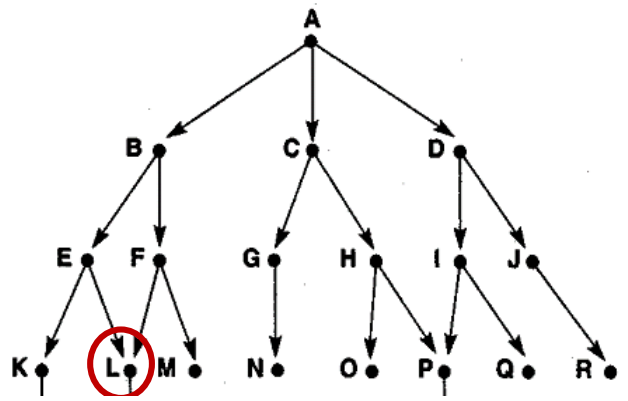
Iterative Deepening Search (IDS) - Example

f = 3

i	[OPEN]	X
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		



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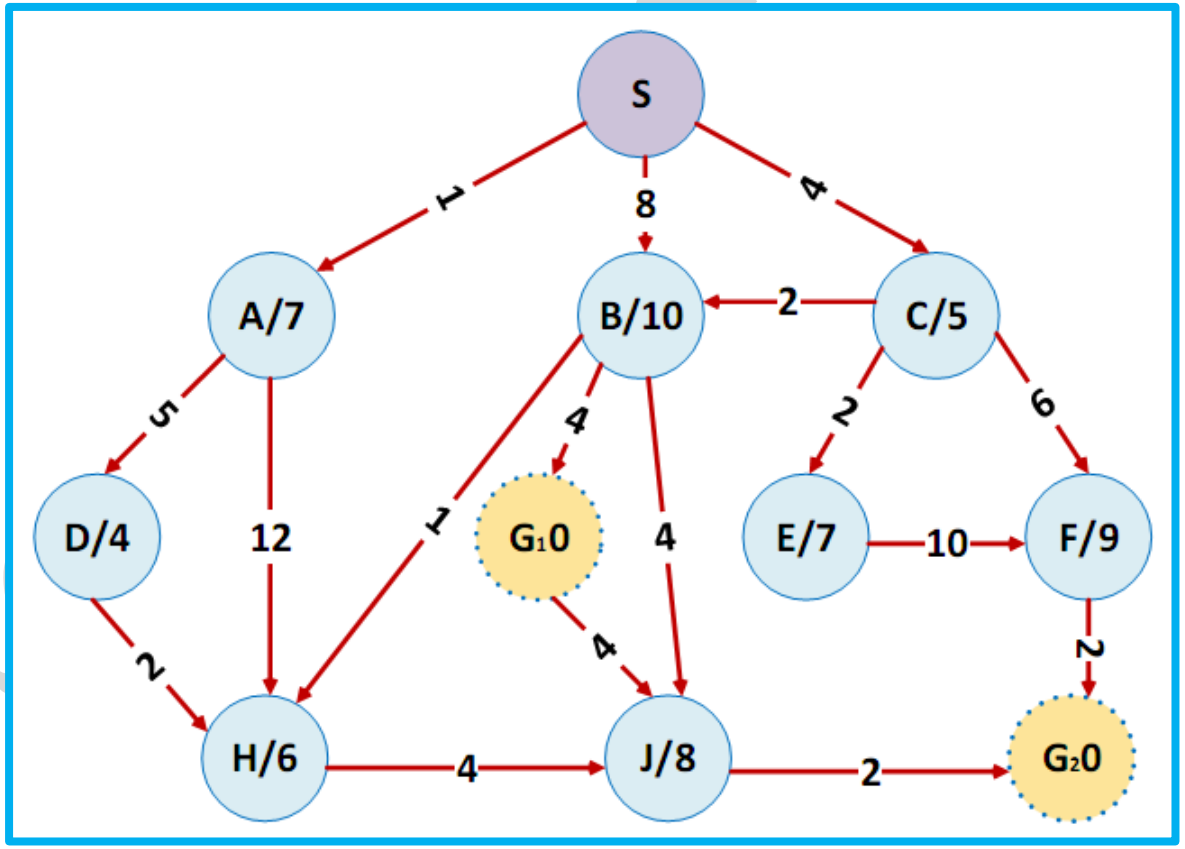
QUIZ

Apply BFS, DFS, DLS and IDS algorithms for the following graph, assume the depth limit (f) is 3.



Quiz Solution

i	[OPEN]	X
1		
2		
3		
4		
5		
6		
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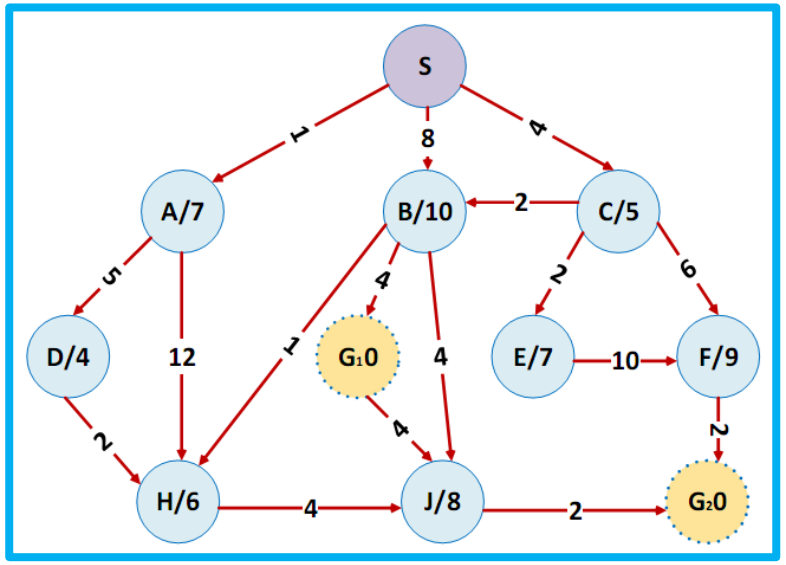
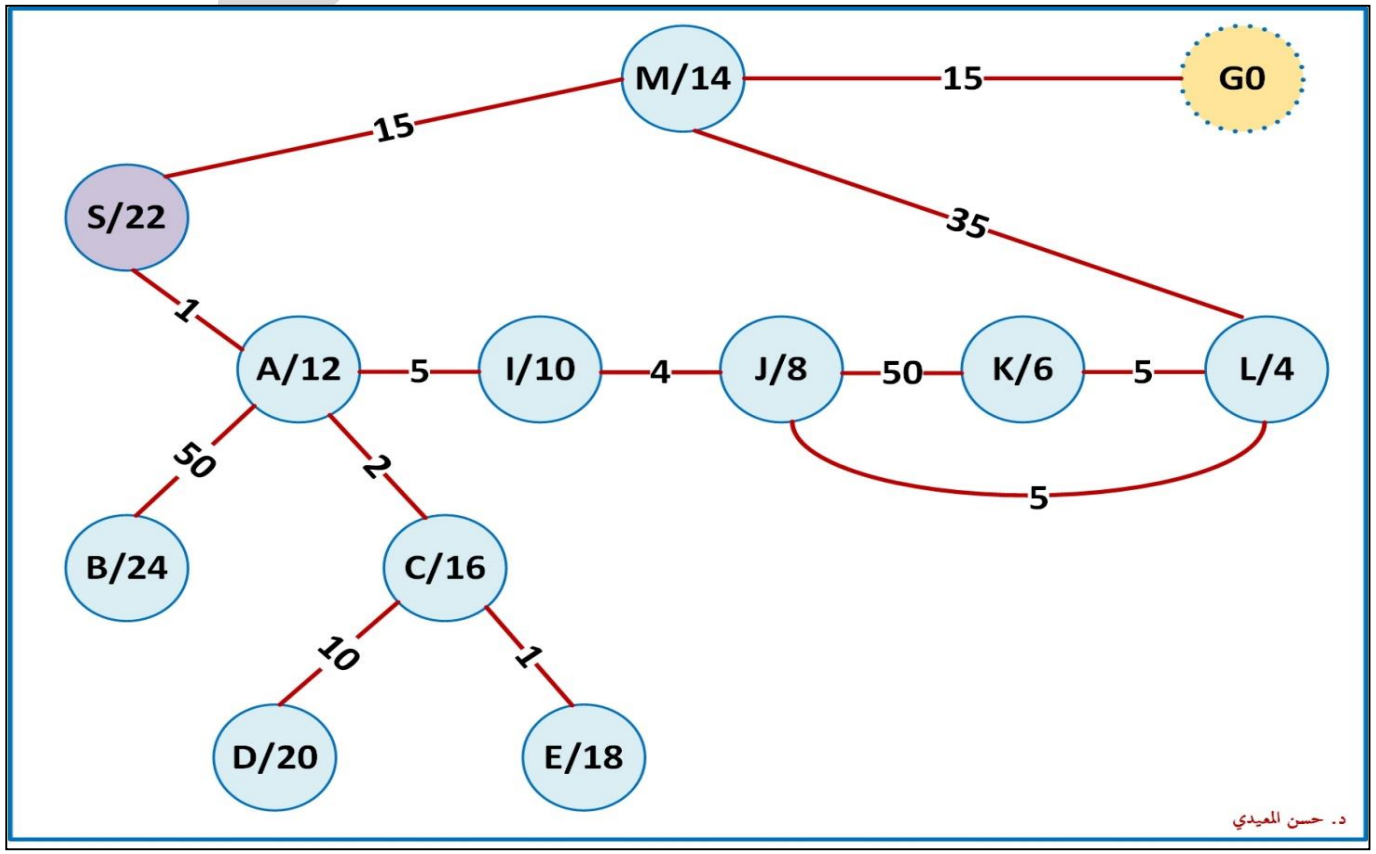
9		
10		
11		



د. حسن المعيدي

QUIZ

Apply BFS, DFS, DLS and IDS algorithms for the following graph, assume the depth limit (f) is 3.



المعيدي

Quiz Solution

i	[OPEN]	X
1		
2		
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7		
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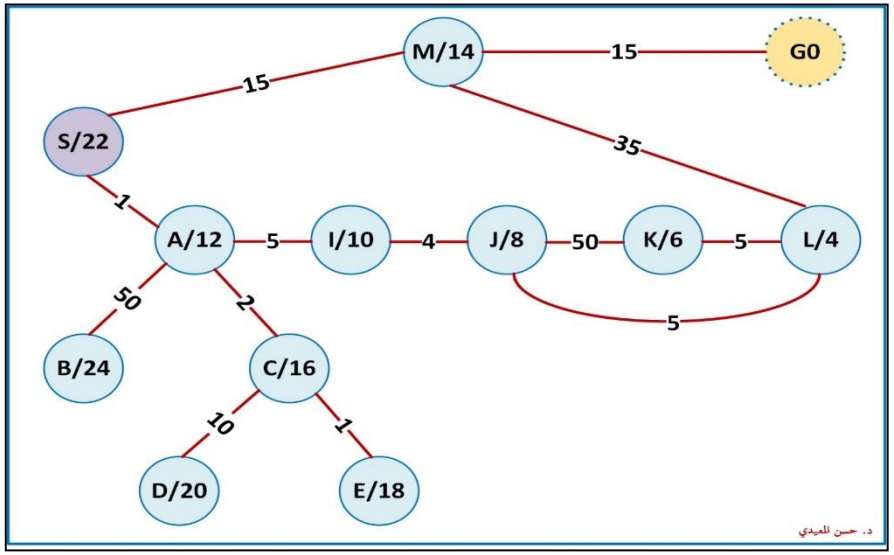
10		
11		



د. حسن المعيدي

Check - Chapter Learning Outcomes

CLO

5

- Define a search problem.
- Compute the size of a search tree.
- Compute the branch factor of a search tree.
- Know the branching factors for some games.
- Compute the number of nodes in a search tree.
- Distinguish between maximum depth and goal depth.
- Approximate a time to search for a goal in a search tree.
- Approximate a space when searching for a goal in a search tree.
- Know different search strategies.
- Understand the criteria for evaluating search strategies.
- Understand, apply and evaluate (BFS, DFS, DLS, and IDS) algorithms.

End of Chapter

5